

Answer Key — Biology Grade 11

Sub-Strand 3.1: Reproduction in Animals • Lesson 6: Helping Where Nature Needs a Hand: Debating IVF, Surrogacy and Sperm Banking

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
C	B	D	C	B	B	C	A

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share on where fertilisation happens in IVF and where the baby develops

Q1. Many people believe that a baby born through IVF grows entirely in a test tube. Which statement about IVF is correct?

- A. Fertilisation happens in the fallopian tube, and only the sperm is prepared in the laboratory
- B. Fertilisation and the whole development of the baby happen in a laboratory dish until birth
- C. Fertilisation happens in a dish in the laboratory, and the embryo is then placed in the uterus where it develops until birth**
- D. Fertilisation happens in the uterus, and the baby is then moved to a laboratory incubator to grow

Correct answer: C. IVF replaces only fertilisation and the first few days of embryo growth outside the body. The embryo is transferred to the uterus, where implantation and pregnancy continue.

USE AFTER: EXPLAIN PHASE — Explain: comparing IVF with external fertilisation in frogs and internal fertilisation in cows

Q2. A frog releases thousands of eggs into water, while a cow produces one calf after months inside her body. Which statement correctly describes how IVF fertilisation compares with these natural methods?

- A. IVF fertilisation and development are both inside the body like in cows, so nothing is done in a laboratory
- B. IVF fertilisation is outside the body like in frogs, but the embryo then develops inside the uterus like in cows**
- C. IVF fertilisation and development are both outside the body like in frogs
- D. IVF fertilisation is inside the body like in cows, and the embryo then develops in water like in frogs

Correct answer: B. In IVF, eggs and sperm meet in a dish (similar to external fertilisation in frogs), but the embryo is transferred to the uterus to develop, as in internal development in cows and humans.

USE AFTER: OBSERVE PHASE — Observe: Case study A (Nakuru, blocked fallopian tubes) in the evidence table

Q3. A woman in Nakuru has blocked fallopian tubes, so her eggs cannot meet sperm. Which technology best addresses her problem?

- A. Taking FSH injections only, so that the tubes open
- B. Sperm banking, where sperm is frozen and stored
- C. Surrogacy only, because her uterus cannot work
- D. IVF, where fertilisation takes place in a dish outside the body**

Correct answer: D. IVF bypasses the blocked tubes by fertilising eggs in a dish and then placing the embryo in the uterus. Her uterus is normal, so surrogacy is not needed, and sperm banking does not solve a blocked tube.

USE AFTER: OBSERVE PHASE — Observe: Case study D (Nairobi, cancer treatment and sperm bank)

Q4. A young man in Nairobi is about to start cancer treatment that may damage his testes. He wants to be able to have children later. Which option is the most suitable?

- A. Surrogacy, because a woman carries his sperm
- B. Using IVF now, because it makes sperm inside the testes
- C. Sperm banking, by freezing and storing his sperm before treatment**
- D. Hormone injections of FSH after the treatment is complete

Correct answer: C. Sperm banks freeze and store sperm so it can be used later, for example in IVF, even if the testes are damaged by treatment. IVF does not make sperm and surrogacy is about carrying a pregnancy.

USE AFTER: OBSERVE PHASE — Observe: Case study E (Mombasa, absent uterus, surrogacy)

Q5. A woman in Mombasa was born without a functional uterus but has healthy ovaries. Which technology allows her to have a genetically related child?

- A. FSH injections, which build a new uterus
- B. Surrogacy, where another woman carries the embryo in her uterus**
- C. Sperm banking, where sperm is frozen
- D. Natural fertilisation in the fallopian tubes followed by development in the ovary

Correct answer: B. Without a working uterus, the embryo cannot develop in her body. In surrogacy, an embryo (usually made by IVF) is carried by another woman. Hormones cannot create a uterus.

USE AFTER: EXPLAIN PHASE — Explain: hormonal flowchart of IVF (hormone injections stimulate eggs to mature)

Q6. A woman in Eldoret has irregular ovulation because her body makes too little FSH and LH. In the IVF flowchart, what is the role of FSH injections?

- A. They freeze the sperm for storage
- B. They stimulate several eggs to mature in the ovaries so the eggs can be collected**
- C. They make the embryo implant without a uterus
- D. They carry the embryo into the uterus

Correct answer: B. FSH stimulates the ovaries to mature several eggs. These eggs are then collected for fertilisation in a dish.

USE AFTER: EXPLAIN PHASE — Explain: building the IVF concept map or hormonal flowchart in pairs

Q7. Which list shows the correct order of the main steps of IVF?

- A. Egg collection, implantation, fertilisation in a dish, hormone stimulation, embryo transfer to the uterus
- B. Embryo transfer to the uterus, egg collection, fertilisation in a dish, hormone stimulation, implantation
- C. Hormone stimulation of eggs, egg collection, fertilisation in a dish, embryo transfer to the uterus, implantation**
- D. Fertilisation in a dish, hormone stimulation of eggs, implantation, egg collection, embryo transfer

Correct answer: C. Eggs must first be stimulated and collected, then fertilised in a dish. The early embryo is transferred to the uterus, where implantation follows.

USE AFTER: END OF LESSON — End: debate and explain-phase discussion correcting myths about infertility

Q8. During the class debate, a learner says: "If a couple cannot have a child, it must be the woman's fault." Which response is scientifically correct?

- A. Infertility can be caused by problems in the man, the woman, or both, and male factors such as low sperm count account for many cases**
- B. Infertility in men never happens because sperm is made in large numbers
- C. Infertility is caused only by cultural beliefs and not by body problems
- D. Infertility is only caused by the woman, because only she carries the baby

Correct answer: A. The lesson corrects the myth that infertility is always the woman's problem. Male causes, such as low sperm count, account for a large share of cases.

